WORKED SOLUTION

STEP I 1991, Question 12

Mechanics • Equilibrium with Friction

STEP I 1991, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q12

The key idea

The plank AB is light and is acted on only at its two ends, by smooth hinges. So the force it exerts on CD must act along AB. That reduces the problem to a single rigid body, CD, with three unknown forces. Taking moments about C gives the thrust in AB, and resolving then gives the friction and the normal reaction at C. The geometry (both planks meet at B) links x to α and β, and this is what makes l and x drop out of the answer.

Solution

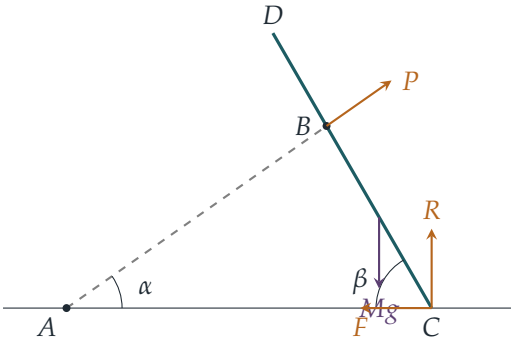
The model

We treat the planks as light rigid rods, the painter as a particle of mass M, and the hinges at A and B as smooth. The ground is rough at C, with coefficient μ.

? **Ask yourself** Why must the force between the planks at B act along AB?

Plank AB has no weight, and the only forces on it come from the smooth hinges at A and B. Taking moments about A, the force at B has no moment about A, so it acts along BA. The force at A must then be equal and opposite to it. By Newton's third law, AB pushes on CD at B with a force P along AB. We'll take P in the direction from A to B (a thrust), and the working confirms that $P > 0$.

So the forces on CD are: the thrust P at B, at angle α above the horizontal and pointing away from A; the weight Mg of the painter, at distance x from C along the plank; and at C a normal reaction R upwards and friction F towards A. The friction has to point towards A because the thrust pushes C away from A.



Geometry

The point B is at height $2l\sin\alpha$, since it is the top of AB. It is also at distance $2x$ from C along CD, so it is at height $2x\sin\beta$. Hence

$$x\sin\beta = l\sin\alpha, \qquad \text{that is,} \qquad x = \frac{l\sin\alpha}{\sin\beta}.$$

The horizontal distance AC is the sum of the horizontal projections of AB and BC:

$$AC = 2l\cos\alpha + 2x\cos\beta.$$

Moments and resolving for CD

Take moments about C. The line of action of P is the line AB, and its perpendicular distance from C is $AC\sin\alpha$. The weight acts at horizontal distance $x\cos\beta$ from C. The two moments are in opposite senses, so

$$P\cdot AC\sin\alpha = Mg\,x\cos\beta \quad\Longrightarrow\quad P = \frac{Mg\,x\cos\beta}{(2l\cos\alpha + 2x\cos\beta)\sin\alpha} > 0.$$

Resolving horizontally and vertically:

$$F = P\cos\alpha, \qquad R = Mg - P\sin\alpha.$$

Substituting for P,

$$R = Mg\left(1 - \frac{x\cos\beta}{2l\cos\alpha + 2x\cos\beta}\right) = Mg\,\frac{2l\cos\alpha + x\cos\beta}{2l\cos\alpha + 2x\cos\beta},$$
$$F = Mg\,\frac{x\cos\beta\cos\alpha}{(2l\cos\alpha + 2x\cos\beta)\sin\alpha}.$$

In particular $R > 0$, so the end C stays in contact with the ground. Dividing,

$$\frac{F}{R} = \frac{x\cos\alpha\cos\beta}{\sin\alpha\,(2l\cos\alpha + x\cos\beta)}.$$

Now we use the geometry, $x = l\sin\alpha/\sin\beta$. Then $x\cos\beta = l\sin\alpha\cot\beta$, and

$$\frac{F}{R} = \frac{l\sin\alpha\cot\beta\cos\alpha}{\sin\alpha\,(2l\cos\alpha + l\sin\alpha\cot\beta)} = \frac{\cos\alpha\cot\beta}{2\cos\alpha + \sin\alpha\cot\beta} = \frac{\cot\alpha\cot\beta}{2\cot\alpha + \cot\beta},$$

where in the last step we divided the top and bottom by $\sin\alpha$.

The plank CD is in equilibrium only if friction can supply this force, that is, $F \leq \mu R$. Since $R > 0$, this is

$$\mu \geq \frac{\cot\alpha\cot\beta}{2\cot\alpha + \cot\beta},$$

as required. (The plank AB is hinged at A, so it is held in place whatever force it needs.)

The final part

If B coincides with D, then $2x = 2l$, so $x = l$. The geometry gives $\sin\beta = \sin\alpha$, and both angles are acute, so $\beta = \alpha$. The planks and the ground form an isosceles triangle with $AC = 4l\cos\alpha$. The friction condition becomes

$$\mu \geq \frac{\cot^2\alpha}{3\cot\alpha} = \tfrac{1}{3}\cot\alpha, \qquad \text{that is,} \qquad \tan\alpha \geq \frac{1}{3\mu}.$$

So equilibrium is possible exactly when $\alpha \geq \alpha_0$, where $\tan\alpha_0 = 1/(3\mu)$.

Ask yourself Where is the painter, and which way does the distance change as α varies?

The painter stands at the midpoint of CD, at distance l from C. Taking A as the origin, C is at $(4l\cos\alpha, 0)$, and the painter is at

$$(4l\cos\alpha - l\cos\alpha,\ l\sin\alpha) = (3l\cos\alpha,\ l\sin\alpha).$$

So the square of the painter's distance from A is

$$d^2 = 9l^2\cos^2\alpha + l^2\sin^2\alpha = l^2\left(1 + 8\cos^2\alpha\right).$$

This decreases as α increases on $(0, \frac{\pi}{2})$. So the largest distance occurs at the smallest allowed angle, $\alpha = \alpha_0$. There $\tan\alpha_0 = 1/(3\mu)$, so

$$\cos^2\alpha_0 = \frac{1}{1+\tan^2\alpha_0} = \frac{9\mu^2}{1+9\mu^2},$$

and

$$d^2_{\max} = l^2\left(1 + \frac{72\mu^2}{1+9\mu^2}\right) = l^2\,\frac{1+81\mu^2}{1+9\mu^2}.$$

Therefore the maximum distance of the painter from A is

$$l\sqrt{\frac{1+81\mu^2}{1+9\mu^2}},$$

as required. This value is attained, with friction exactly limiting.

Checks

- Limiting cases of the final answer: as $\mu \to 0$ the distance tends to l, which corresponds to $\alpha \to \frac{\pi}{2}$, with both planks vertical. As $\mu \to \infty$ it tends to $3l$, which corresponds to the planks lying flat with the painter at $(3l, 0)$. Both are sensible.
- If $\beta \to \frac{\pi}{2}$, so that CD is vertical, then $\cot\beta \to 0$ and the required μ tends to 0. A vertical plank needs no friction, which agrees.
- Numerically, taking $\alpha = 60°$ and $\beta = 70°$ and solving the three equilibrium equations for CD directly gives $F/R = 0.13837$, which matches the formula. A scan over α with $\mu = 0.5$ gives a maximum distance of $2.55704\,l$, and the formula gives $l\sqrt{21.25/3.25} = 2.55704\,l$.

Common slips

- Assuming that the force at B is horizontal or vertical. Because AB is light and smoothly hinged at both ends, the force must act along AB.
- Forgetting the vertical component of the thrust in R. It reduces the normal reaction, which makes friction harder to supply.
- In the final part, maximising over all α instead of only over the angles for which equilibrium is possible.

Beyond STEP

A light rod with smooth pins at its ends is a *two-force member*. Recognising these is the first step in analysing pin-jointed frameworks (trusses) in engineering statics, where each such member carries a pure tension or thrust. The final part is a small constrained optimisation: the objective is monotonic, so the optimum sits on the boundary of the feasible set. This is a theme that recurs throughout linear programming and optimisation. For more on equilibrium of rigid bodies, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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